

Ethics in Practice: Empirical and Technical Design for Safer Social VR - Insights from the META-TOO EU project

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ABSTRACT

Social VR is an immersive virtual realm where individuals can socialize, work, and entertain. Platforms like Horizon Worlds or VR Chat enable users to be immersed in the virtual environment thanks to the inclusion of multisensory channels and embodiment into a virtual representation of themselves (avatar). The combination of an open social VR platform which offers obscured identity, synchronicity, and ephemerality, paired with a deep sense of embodiment and presence gives way to cyber-harassment. Verbal, physical, and environmental abuse intensifies emotional impact, especially for users with self-similar avatars, marginalized identities and vulnerable population, such as children and individuals with mental health difficulties. There is a lack of understanding of the factors which influence harassment and inappropriate social behaviour in Social VR. The EU funded META-TOO project addresses this gap through two experimental tracks, first one focused on behavioral and emotional impact studies, and second on the investigation of safety tools which can be used by users to mitigate the effects of harassment. In this paper, we present the two experimental frameworks of META-TOO and discuss the associated ethical challenges when conducting this type of research.

Index Terms: Social VR, harassment, experimental design, ethics

1 INTRODUCTION

Harassment in social VR is routinely reported as more intense and “real” than in-text or screen-based platforms because of synchronous voice, full or partial body tracking, and the sense of physical copresence [3, 7, 17]. Users describe verbal, physical, and environmental harassment (e.g., groping, stalking, surrounding avatars, manipulating the environment) that can feel analogous to offline assault, particularly when perpetrated against self-similar or identity-expressive avatars [3, 7, 20, 19]. Women and other marginalized users/participants report disproportionate targeting and must frequently trade off authenticity of self-representation against safety, for example by altering avatar design or voice use to avoid sexualized or identity-based harassment [14, 15, 20]. Bystanders, not only direct targets, report distress and discomfort when witnessing harassment in social VR spaces [3, 7, 13, 11], underscoring the need

to treat harm as relational and environmental rather than purely individual.

Immersive VR also raises specific mental health concerns. Stress-inducing VR paradigms can reliably elicit negative affect, elevated heart rate, and intrusion/avoidance-like symptoms, even with comparatively “mild” scenes [6]. A scoping review of sexual violence in VR highlights risks of retraumatization, depersonalization, and disorientation, while also noting a lack of standardized outcome measures and ethical guidelines for simulations involving sexual violence and harassment [13]. At the same time, controlled VR has been used therapeutically for post traumatic stress disorder (PTSD) and anxiety, exploiting the capacity for graded exposure in safe, controllable environments [6, 1]. This ambivalence reinforces the call by Slater and others for empirical measurement of presence, embodiment, and behavioral impact as basic prerequisites for ethical VR design, and for treating psychological safety and autonomy as public goods to be monitored, not incidental byproducts [11, 8].

2 ETHICAL AND METHODOLOGICAL DESIGN IN META-TOO

The META-TOO project responds to these challenges through two complementary tracks:

- **Behavioral & Emotional Impact Studies.** Building on controlled harassment and street harassment simulations [19, 9, 18], META-TOO develops standardized scenarios that vary the degree of avatar self-similarity [19], modality and intensity of harassment (verbal, physical, environmental) [3, 20, 5], social configuration (presence of bystanders) [7, 11, 9], AI involvement in the experimental scenario.
- **Safety Mechanism Prototyping.** Empirical work in social VR shows that current tools (blocking, muting, reporting, personal space boundaries) are often poorly understood, inconsistently available across platforms, and hard to access in the moment of harassment [3, 7, 10]. META-TOO tests: configurable personal space “bubbles” and collision constraints, consent-based interaction mechanics (opt-in touch, teleports, proximity) [15, 10], graded sanctions and “house rules” enforced through warnings, temporary restrictions, and license revocation [3, 10], human-AI collaborative moderation, including AI support for triaging reports and detecting highrisk patterns [15, 12].

3 CHALLENGES IN DESIGNING HARASSMENT EXPERIMENTS IN VR

Immersive social VR can heighten emotional impact through embodiment, social presence, and synchronicity, but these same properties complicate ethical control and experimental validity [13], as reflected in the META-TOO project’s recurring challenges for both participant safety and empirical interpretation. Ecological validity often relies on realistic, emotionally intense harassment scenarios, yet increasing realism elevates psychological risk, prompting conservative design constraints that may weaken presence and engagement and thus create tension between scientific value and protection from distress or retraumatization [13, 9, 6]. Similarly, embodiment

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is widely used to foster perspective-taking, but without meaningful agency participants may feel constrained or unable to intervene, and meta-analytic and review work shows that immersive interventions often yield mixed or null effects on explicit attitudes, underscoring the importance of context and interaction [11]. Design choices such as scripting and replay can secure experimental control but risk breaking presence and plausibility when events do not respond contingently to the user, undermining the very mechanisms embodiment-based interventions seek to harness [21]. Finally, reliance on explicit self-report in populations with low baseline endorsement of harassment can produce ceiling effects that obscure more subtle changes in sensitivity, emotion, or behaviour, meaning that absent attitude shifts should not be taken as absent impact and that systematic evaluation of experiential qualities such as presence, plausibility, agency, and social contingency is crucial when designing harassment experiments in Social VR [11, 18].

4 CORE ETHICAL QUESTIONS AND CONCRETE DESIGN RESPONSES

Technical Integration combines VR platform development for social interaction simulation with psychological evaluation and ethical protocols must be implemented during experiments to take into account some key ethical questions:

- Is it ethical to run experiments involving harassment remotely (e.g., on a social VR platform) for higher ecological validity and easier recruitment of actual VR users?
- Where and how to recruit participants, who should be the target population? How can we ensure that participants will not be re-traumatized (as in: should we find participants who were not abuse survivors?)
- How can we induce stress and how to measure it? What are the important factors to consider?
- Is it ethical to use AI and open scenarios for improved ecological validity of the experience? What criteria should we use to choose representative avatars for embodiment and body swapping exercises, and how can we avoid stereotypical or harmful representations?
- How should we inform participants about the purpose of the study, the nature of avatars, and the potential risks of being exposed to cyber-bullying interactions?
- What details should we include in the informed consent form to ensure that participants understand the potential risks and benefits?
- How can we minimise psychological risks for participants, especially those who may be more sensitive to the experience of cyberbullying? What support measures should we put in place for participants after the experience (e.g. psychological support resources, reporting mechanisms)?

4.1 Target population selection and recruitment for measuring stress and defining ethical limits

Self-similar avatars can deepen embodiment and thereby magnify emotional impact of harassment [3, 19, 8]. Meanwhile, avatar appearance and anthropomorphism shape perceived harassment severity and user responses[20]. Ethically responsible avatar design in META-TOO therefore requires: (i) Systematic variation of self-similar vs stylized avatars to quantify risk amplification, with conservative defaults for first-time or vulnerable users [19, 8]; (ii) Avoidance of stereotypical racialized, gendered, or sexualized body

types, building on critiques that current Social VR design often reproduces offline cisnormativity, male-dominated power structures [14, 15, 8]; (iii) Inclusion of diverse, non-caricatured avatars and options for identity obfuscation where desired, informed by findings that identity revelation and misperception strongly shape identity-based harassment risk [15].

Thus, the challenges related to studying cyberharassment with embodied avatars concentrate in the representation of the Social VR users/participants. The question is how can we represent diversity of the VR community without falling into stereotypes and clichés? Are realistic humanoid avatars preferred or should one replicate avatars often seen on VR platforms, such as animal-like or cartoonish avatars?

4.2 Informed consent

Ethical research on harassment in immersive VR requires forms of informed consent that go beyond standard online or screen-based studies. Recent work in HCI has shown that consent practices are highly heterogeneous and often poorly adapted to complex, embodied, or emotionally sensitive study designs. Consent procedures frequently rely on generic templates, rarely assess participant understanding, and tend to under-specify experiential risks, especially in studies involving immersive exposure, deception, or affective manipulation [16]. These limitations are especially consequential in social VR harassment research, where participation entails not only data collection but also embodied exposure, identity representation, and the possibility of emotional harm unfolding dynamically during interaction.

At a minimum, informed consent in this context should include a plain-language description of the presence of simulated harassment or cyberbullying, the range of potential emotional reactions (e.g., distress, anger, dissociation, intrusive memories), and the role of avatars in the study, including degrees of self-similarity and the presence of possible identity cues [13, 19, 9, 6]. Participants should be explicitly informed that they may stop or withdraw at any time without penalty, supported by clear, in-scenario stop affordances mapped to familiar controller actions rather than abstract instructions [13, 6]. Where appropriate, particularly for minors or novice VR users, brief comprehension checks prior to participation can help ensure that consent is not merely formal but meaningfully understood [13, 18].

Consent forms should also clearly describe data practices around biosignals, behavioral logs, and interaction traces, given the sensitivity of stress- and harassment-related data and the potential for future profiling or misuse [15, 5, 12]. At the same time, an unresolved ethical tension remains: researchers must prepare participants for potentially distressing experiences without disclosing so much detail that it induces demand characteristics or constrains naturalistic behavior. Navigating this tension remains a central challenge for ethical experimentation in social VR.

4.3 Minimizing psychological risk and avoiding victim blaming

Empirical work on VR simulations of harassment and prejudice shows that even well-intentioned embodiment interventions can still be harmful or ethically ambiguous, particularly when they intensify exposure without offering agency or meaningful avenues for action [13, 11, 9, 8]. Ethical design in this space therefore requires not only mitigation of distress, but also careful attention to how responsibility, agency, and harm are framed within the experimental scenario.

To reduce psychological risk and avoid victim-blaming dynamics, several design principles are essential. First, scenarios should be framed around system-level factors and perpetrator or bystander behavior (e.g., how platforms respond, how norms are enforced, how bystanders intervene), rather than positioning participants as

passive recipients who must endure abuse or learn to tolerate it [13, 9, 8].

Second, structured debriefings should be integral to the study design, explicitly normalizing distress responses, situating harassment as a structural and social problem rather than an individual failure, and directing participants to coping strategies and reporting mechanisms [13, 9, 6]. Third, outcome measures should extend beyond immediate affect to include validated scales of sexism, moral disengagement, and bystander intention, ensuring that interventions reduce tolerance of harassment and do not merely shift the burden of prevention onto targets [18].

Despite these safeguards, a fundamental challenge in this area lies in experimental manipulation. One may ask whether certain studies should not be conducted at all in the name of protection of participants and where is the line that shall not be crossed? How can one ethically use AI in scenario building? Is it acceptable to use AI for enhanced realism and interactivity of the scenarios, considering that it may lead to unpredictable dialogue outcomes? In this sense, minimizing harm in social VR harassment research requires not only technical controls and post-hoc support, but also principled decisions about where the line between acceptable exposure and excessive risk should be drawn.

4.4 Post-experience support and reporting

Best practice emerging from trauma-analog VR and sexual violence simulations includes: (i) immediate postsession checkins and short questionnaires on affect, presence, and embodiment to detect lingering distress [9, 18, 6]; (ii) provision of psychological support resources (oncampus counselling, helplines, online support) in writing and verbally, regardless of apparent distress level [13, 9, 6] and (iii) simple, in-scenario and post-scenario mechanisms for reporting discomfort about the scenario itself, the experimenters, or other participants, feeding into iterative ethical review [3, 7, 10, 6].

4.5 Best practices associated with the project META-TOO

The above ethical questions are currently being implemented in the experimental part of the META-TOO project. While the experimentation is still ongoing, some specific questions and solutions have been proposed which are associated with the two experimental frameworks. In addition to following the standard principles of participant protection, we also implemented:

Ethics Advisory Board (EAB). While each partner in the META-TOO project is associated with their own institutional ethics procedures, a multidisciplinary independent EAB was appointed as part of the META-TOO project, featuring experts already engaged in addressing these ethical issues at national and EU levels: computer scientists, psychologists and philosophers, as well as jurists in digital law and European AI governance. Discussions with the EAB on the results of our studies lead to two conclusions: 1) the EAB specifically helps to address the specific ethical challenges of our project, and 2) it offers a broader view of the ethical dimension encouraging us to organise a full workshop dedicated to the topic (upcoming).

Community engagement. The creation of virtual humans which represent harassers and victims is limited to already available materials, but a considerable effort is being made to not create offensive, stereotypical or simply non-representative characters. To solve this problem, we consulted the agent design with the representative minority groups. For example, when creating non-binary virtual characters, we have collaborated with several nonbinary people to evaluate and adjust the characters to be representative. Care is also given to provide enough variation when creating the ‘harasser’ in our scenarios, to not always represent one type of character.

Safety tool taxonomy. Because social VR is a relatively new social media platform, most of the safety feature designs have been

imported from other social media which led to inconsistent terminology we have today. For this reason, we have developed a taxonomy which classifies existing tools into categories, created based on existing theories of social interaction. We also put special importance on the process of activating the safety tool, as this is important for its effectiveness [2]. To ensure a structured approach to creating relevant and effective safety tools, we designed our own safety tools based on this taxonomy.

Future designs. In the experiment designs in META-TOO, agents are used in the role of the ‘harasser’ to avoid assigning potentially harming roles to participants. There is an ongoing discussion on whether integrating LLM functionality and other AI-based interaction elements would raise significant concern in terms of privacy, controllability and behaviour validity. There have been many initiatives to improve the safety of the AI tools, such as explainable AI, but these solutions are still being investigated.

Another pending question is incorporating wider cultural dimension in the SVR design. In the META-TOO project, we are exploring the mitigation of gender-based harassment in SVR but the design of this digital media itself could be catered more towards male users, for example. We also do not include children in our experimental framework even though they are the most vulnerable users in SVR.

5 CONCLUSION: TOWARD AN ETHICS OF REPRESENTATION AND ACTION IN SOCIAL VR

Ethical analysis of VR stresses two intertwined dimensions: what kinds of actions are made possible, and how people and worlds are represented [4]. In social VR harassment research, this translates into: (i) Careful constraint of possible harmful actions in experimental platforms (e.g., limits on proximity, touching, disrobing, or degrading animations), even if such actions exist “in the wild” [14, 3, 20, 4]; (ii) Ongoing critical review of representation: whose bodies and identities are visible, who is positioned as victim, observer, or agent of change, and whether scenarios reproduce existing inequities [14, 15, 4, 8].

By combining controlled empirical work on stress, embodiment, and harassment dynamics with iterative technical design of safety features and robust ethical protocols, projects like META-TOO can help move social VR from reactive moderation toward proactive, evidence-based protection of mental health, dignity, and autonomy in immersive social spaces. At the same time, our experience in META-TOO underscores the need for explicit, community-level discussion about the limits of experimentation in social VR. As immersive systems increasingly enable realistic harassment, identity targeting, and AI-driven interaction, not all technically feasible studies should be considered ethically permissible. We argue for shared tools and guidelines around upper bounds of induced distress, acceptable degrees of deception and loss of agency, and conditions under which ecological validity no longer justifies experimental harm. Advancing ethics in social VR therefore requires not only better tools and protocols, but also collective restraint.

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